

2016 Level II Mock Exam: Afternoon Session

The afternoon session of the 2016 Level II Chartered Financial Analyst® Mock Examination has 60 questions. To best simulate the exam day experience, candidates are advised to allocate an average of 18 minutes per item set (vignette and 6 multiple choice questions) for a total of 180 minutes (3 hours) for this session of the exam.

Questions	Topic	Minutes
1-6	Ethical and Professional Standards	18
7-12	Quantitative Methods	18
13-18	Financial Reporting and Analysis	18
19-24	Financial Reporting and Analysis	18
25-30	Equity	18
31-36	Equity	18
37-42	Fixed Income	18
43-48	Derivatives	18
49-54	Alternative Investments	18
55-60	Portfolio Management	18
	Total:	180

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Ethics - Carlyle

Lindsey Carlyle, CFA is a research analyst with Woodstock Brothers (Woodstock), a division of the Woodstock Group. Woodstock is a full service brokerage and investment firm. The Woodstock Group and all of its divisions have adopted the CFA Institute's Code of Ethics and Standards of Professional Conduct. Carlyle is writing a research report on Paladin Shippers (Paladin), a specialty transportation company. From time to time, several of Woodstock's senior managers have served on the Board of Directors of Paladin, although none of them currently serve on Paladin's Board. Additionally, Woodstock's investment banking division, separated by a Chinese wall from the research division, represented Paladin in their initial public offering several years ago.

Carlyle initiates coverage of Paladin with a "Strong Buy" recommendation for accounts with a high risk tolerance over the next six months. In the report she identifies the main factors she utilized in determining her recommendation. She does not mention the previous relationships Woodstock had with Paladin as they no longer exist. The report was approved by her supervisor before being disseminated to all clients with a high risk profile including Marietta Investments (Marietta) the asset management division of Woodstock Brothers.

Scott Robinson, CFA is a portfolio manager for Marietta. He started at Marietta as a junior analyst working his way up to his current position as a portfolio manager. When he first joined the portfolio management group he worked with the high growth equity team. Robinson now specializes in managing small cap value stocks for U.S. pension funds and a few high net worth individuals. Robinson knew Carlyle from his previous days in the research department. Based only on Carlyle's recommendation, Robinson reviews the accounts under his management and purchases Paladin stock where suitable.

All of Robinson's accounts have given him full discretion in managing their funds. Given his decision to purchase Paladin, he determines the total number of shares that need to be purchased for all of his accounts and submits the block buy order to Marietta's trading department. It takes several hours to fully execute the trades and the shares are allocated according to Marietta's Trade Allocation Policy. The policy states, "Client accounts participating in a block trade shall receive the same execution price and be charged the same commission, if any. All trade allocations shall be made on a pro rata basis prior to or immediately following a partial or complete block trade."

As part of the block trade, Robinson purchases Paladin Shippers for a mutual fund he manages. Reviewing all the trades executed in the mutual fund is the responsibility of Renee Stevens, CFA, Marietta's Chief Compliance Officer. Given Paladin's rich valuation, she is concerned about Robinson's purchase of the stock for the fund and requests he provide justification for the purchase. Robinson gives Stevens the following explanation:

"Subsequent to the "Strong Buy" recommendation from Woodstock Brothers research analyst Lindsey Carlyle, Paladin Shippers was purchased due to its excellent return profile and growth opportunities. While the purchase of what appears to be a large cap growth stock in a small cap value portfolio might

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be questionable, it fits nicely within the overall purpose and objectives of the fund. Additionally, the portfolio characteristics of the fund, even with the inclusion of Paladin, clearly place the fund in the small cap value style category.”

Following the end of month reporting, Robinson received an email from one of his clients questioning the purchase of Paladin for his portfolio. The client felt the purchase was not in compliance with his Investment Policy Statement. Robinson again reviews the client’s investment policy and prepares a response. He also plans to invite the client to an upcoming meeting he hopes to schedule with Carlyle to discuss her recommendation on Paladin.

Carlyle remains impressed with Paladin’s growth opportunities and attention to customer satisfaction. Robinson asks Carlyle to meet with several potential new clients for Marietta. He believes that demonstrating his direct access to Street research analysts will help him gain new clients. Several minutes prior to the start of the meeting Robinson introduces Carlyle to one of his existing clients. Carlyle reviews with them a recent conversation she had with the Paladin CFO regarding new growth opportunities which she plans to include in her next research report. Carlyle meets with Robinson’s potential clients and discusses the stock research process.

-
1. With regards to Carlyle’s research report on Paladin, which of the CFA Institute Standards of Professional Conduct did she *most likely* violate?
- A. Communication with Clients and Prospective Clients
 - B. Disclosure of Conflicts
 - C. Suitability

Answer = B

Woodstock’s previous business relationships needs to be disclosed in Carlyle’s initial research report in order to maintain independence and objectivity. Although no officers are currently serving on the board and a Chinese wall exists, it does not eliminate the need for the client to assess the potential conflict of interest between Woodstock and Paladin and the reason behind Carlyle’s investment recommendation. Standard VI(A)–Disclosure of Conflicts requires charterholders and candidates to disclose all matters that could reasonably be expected to impair their objectivity and ability to judge motives and possible biases. Therefore Carlyle violated Standard VI(A).

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“Guidance to Standards I–VII,” by CFA Institute
Standard VI(A)–Disclosure of Conflicts

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2. By purchasing Paladin stock for his client accounts, did Robinson *most likely* comply with the CFA Institute Standards of Professional Conduct?

- A. No, because he is familiar with Woodstock's research methodology
- B. Yes
- C. No, because he based his purchase on Carlyle's analysis

Answer = C

Robinson relied solely on Carlyle's "Strong Buy" recommendation to make the investment decision to purchase Paladin's stock and thus did not comply with Standard V(A)–Diligence and Reasonable Basis. Even though he is familiar with Carlyle and Woodstock's investment methodology, Standard V(A) requires that he make a reasonable and diligent effort to determine whether her research is sound, her assumptions are reasonable, the analysis is timely, and her recommendation is objective.

CFA Level II

"Guidance to Standards I–VII," by CFA Institute
Standard V(A)–Diligence and Reasonable Basis

3. Does the manner in which Robinson allocates the purchase of Paladin shares for the accounts under his management *most likely* violate the CFA Institute Standards of Professional Conduct?
- A. No.
 - B. Yes, because trades should not be allocated until the block is fully traded.
 - C. Yes, Marietta's trading policy does not differentiate between discretionary and non-discretionary accounts.

Answer = A

There is no violation. Standard III(B)–Fair Dealing states that members must deal fairly and objectively with all clients regardless of whether they are discretionary or non-discretionary accounts. Marietta's trade allocation policy treats all clients fairly in terms of both trade execution order and price.

CFA Level II

"Guidance to Standards I–VII", by CFA Institute
Standard III(B)–Fair Dealing

4. Does the inclusion of Paladin's stock in the small-cap value mutual fund *most likely* violate any of the CFA Institute Standards of Professional Conduct?
- A. Yes, Robinson violates with respect to suitability.

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- B. Yes, Stevens violates with respect to her responsibilities to supervise.
- C. No.

Answer = C

No violation has taken place. Standard III(A)—Loyalty, Prudence, and Care states that investment decisions may be judged in the context of the total portfolio rather than by individual investments within the portfolio. Robinson has satisfied his duty by thoroughly considering Paladin's place in the overall portfolio.

CFA Level II

"Guidance to Standards I–VII," by CFA Institute
Standard III(A)—Loyalty, Prudence and Care

5. What is Robinson's *best* action with regard to the email he received from his client about the purchase of Paladin? Robinson should:
- A. advise him that the trades were reviewed by the firm's chief compliance officer.
 - B. ask to meet with him to review and, if necessary, revise his investment policy statement.
 - C. liquidate the shares of Paladin in the client's portfolio.

Answer = B

Standard III(C)—Suitability requires members in an advisory relationship to regularly update the client's investment policy statement to reflect any changes in the client's circumstances. Updating the investment policy statement should be repeated at least annually. Without efforts to update information concerning client factors, one or more factors could change without the investment manager's knowledge.

CFA Level II

"Guidance to Standards I–VII," by CFA Institute
Standard III(C)—Suitability

6. During Carlyle's meeting with Robinson's potential clients, who *most likely* committed a violation of the CFA Institute Standards of Professional Conduct?
- A. Robinson
 - B. Neither Carlyle nor Robinson
 - C. Carlyle

Answer = B

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Neither Carlyle nor Robinson has violated the CFA Institute Standards of Professional Conduct during the meeting with Robinson's potential clients. By asking Carlyle to meet with several potential new clients Robinson is simply attempting to grow his client base. By merely discussing her research approach Carlyle has not given preferential treatment to these potential clients. Carlyle has not violated the standards as she has only discussed the research process and did not disclose her most current research.

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"Guidance to Standards I – VII," by CFA Institute
Standard III(B) Fair Dealing

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Quant - Hahn

Rolf Hahn is an analyst with Weissdorn GmbH. He is puzzled by the most recent report of his firm's statistical analysis of one of its investments, VeriZoom, Inc. The reported correlation of VeriZoom with the market seems to be too high. He suspects that the new software the firm installed the previous week is not programmed to correctly calculate correlations. Hahn decides to double check the values given in the report.

Hahn accesses Weissdorn's returns database and downloads the appropriate monthly returns data needed to estimate the correlation of VeriZoom and the MSCI Europe Large Cap Index, the index Weissdorn uses as the market proxy. Using a spreadsheet program, Hahn establishes the variance and covariance data shown in Exhibit 1.

Exhibit 1		
Variance and Covariance Data Related to VeriZoom		
	Variance of Returns	Covariance with the MSCI Europe Large Cap Index
VeriZoom	0.0225	0.022
MSCI Europe Large Cap Index	0.04	0.04

Given the results in Exhibit 1, Hahn's suspicion that the software is incorrect with the correlations is confirmed. He asks his assistant, Angela Greene, to help him recalculate the statistics for each firm Weissdorn covers. While working to complete this task, Greene asks Hahn why correlation is so important in portfolio management. Hahn replies that correlation can establish the strength of the relationship between two variables. He states that correlations close to;

- 1 +1 indicate there is a strong direct relationship between the variables,
- 2 0 indicate there is no relationship between the variables, and
- 3 -1 indicate there is a strong offsetting relationship between the variables.

Greene responds by asking if the only means of evaluating whether the correlation is close to 0 is the judgment of the analyst. Hahn says that there is a statistical test to determine whether a sample correlation is statistically significantly different from 0. Hahn takes the results for another firm that Weissdorn follows, Anchor-Wise, Inc. (AWI), and uses them to demonstrate the test. The data for Anchor-Wise are shown in Exhibit 2.

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Exhibit 2 Correlation of Anchor-Wise with MSCI Europe Large Cap Index		
Correlation	Number of Observations	Critical Value at 5% Level of Significance
0.267	60	1.67

Greene asks Hahn if there are other techniques that can be used to evaluate the relationship between two variables. Hahn answers that regression, a technique closely related to correlation analysis, can be used to establish the relationship between two variables within the assumptions of the analysis.

For example, he says that they might want to evaluate the following:

$$R_{AWI} = b_0 + b_1 R_{MSCI} + \epsilon$$

where R_{AWI} is the monthly total return on AWI, R_{MSCI} is the monthly total return on the MSCI Europe Large Cap Index, and ϵ represents the error term. Greene and Hahn use the data they downloaded for AWI and the MSCI index and run the regression. The results are given in Exhibit 3.

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Exhibit 3 Results of a Simple Regression $R_{AWI} = b_0 + b_1 R_{MSCI} + \epsilon$			
Regression Statistics			
Multiple R	0.2666		
R^2	0.0711		
Standard error of estimate	0.0343		
Observations	60		
Analysis of Variance	Degrees of Freedom (df)	Sum of Squares (SS)	
Regression	1	0.0052	
Residual	58	0.0685	
Total	59	0.0738	
	Coefficients	Standard Error	t-Statistic
b_0	0.005	0.0045	1.1151
b_1	0.0634	0.0301	2.1069

Hahn next considers the topic of multiple regression. To provide an example that he and Greene can review, he downloads data that allow him to establish the following:

$$R_{AWI} = b_0 + b_1 \Delta GDP_{USA} + b_2 YC_{USA} + \epsilon.$$

where R_{AWI} is the quarterly return for AWI, ΔGDP_{USA} is the change in US real GDP for the quarter, and YC_{USA} is the slope of the yield curve (measured as the difference between the 1-month US Treasury rate and the 10-year Treasury rate) at the end of the quarter. The results of the regression are reported in Exhibit 4.

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Exhibit 4 Results of a Multiple Regression $R_{AWT} = b_0 + b_1 \Delta GDP_{USA} + b_2 YC_{USA} + \epsilon.$			
Regression Statistics			
Multiple R	0.096		
R ²	0.0092		
Standard error of estimate	0.0303		
Observations	39		
Analysis of Variance	Degrees of Freedom (df)	Sum of Squares (SS)	
Regression	2	0.0003	
Residual	36	0.0009	
Total	38	0.0334	
	Coefficients	Standard Error	p-Value
b_0	0.0014	0.0121	0.9089
b_1	0.0001	0.0028	0.971
b_2	0.0025	0.0045	0.5849

Hahn points out to Greene that interpretation of the results from a multiple regression can be biased unless various conditions are met. He mentions three such conditions. The error term must

- have an expected value that is less than zero,
- be strongly positively correlated across observations, and
- be normally distributed.

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7. Based on the data in Exhibit 1, the correlation between VeriZoom and the MSCI Europe Large Cap Index is *closest* to:

- A. 0.980.
- B. 0.550.
- C. 0.733.

Answer = C

The correlation is calculated as

$$r = \frac{\text{Cov}(X, Y)}{s_X s_Y},$$

$$\text{In this case, } r = \frac{0.0220}{\sqrt{0.0225} \times \sqrt{0.0400}} = 0.7333.$$

CFA Level II

"Correlation and Regression," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle
Section 2.3

8. Which of Hahn's three statements regarding correlation is *least accurate*?

- A. Statement 3
- B. Statement 2
- C. Statement 1

Answer = B

Correlation measures the linear association between two variables. Variables with a correlation close to 0 can nonetheless exhibit a strong relationship—just not a linear relationship.

CFA Level II

"Correlation and Regression," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle
Section 2.4

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9. Assuming that the returns for Anchor-Wise and the returns for the MSCI Europe Large Cap Index come from a bivariate normal distribution, the results shown in Exhibit 2 *most likely* indicate that:
- A. their correlation is statistically significantly different from 0.
 - B. there is a non-linear relationship between the variables.
 - C. their correlation is not statistically significantly different from 0.

Answer = A

The test statistic, t , is: $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}},$	$t = \frac{0.267\sqrt{60-2}}{\sqrt{1-0.267^2}} = \mathbf{2.110}$
where r is the correlation and n is the number of observations.	Because 2.110 is greater than the critical value of 1.670, we reject the null hypothesis that the correlation is 0.

CFA Level II

"Correlation and Regression," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle
Section 2.6

10. Based on the results reported in Exhibit 3, the F -statistic to test whether the slope of the regression line is different from zero is *closest* to:
- A. 4.157.
 - B. 2.107.
 - C. 4.403.

Answer = C

For a regression with one independent variable, as is the case here, the F -statistic is calculated as

$$F = \frac{\frac{RSS}{SSE}}{\frac{n-2}{58}} = \frac{0.0052}{0.0685} = 4.403,$$

where RSS is the regression sum of squares, SSE is the residual sum of squares, and n is the number of observations.

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"Correlation and Regression," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle
Section 3.6

11. At the 5% level of significance, Exhibit 4 indicates that:

- A. only b_1 is statistically significantly different from zero.
- B. none of the coefficients in the regression are significantly different from zero.
- C. b_0 and b_1 are both statistically significantly different from zero, but b_2 is not significant.

Answer = B

All three coefficients are not statistically significantly different from zero at the 5% level; their p -values are too high. To accept the hypotheses that they are statistically different from zero, the reported p -value needs to be equal to or less than 0.05.

CFA Level II

"Multiple Regression and Issues in Regression Analysis," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle
Section 2

"Correlation and Regression," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle
Section 3.5

12. Which of Hahn's conditions relating to the error term in a multiple regression is correctly stated?

- A. The condition that relates to the expected value
- B. The condition that relates to the correlation
- C. The condition that relates to the distribution

Answer = C

Multiple linear regression assumes that the error term is normally distributed, has an expected value of zero, and is uncorrelated across observations.

CFA Level II

"Multiple Regression and Issues in Regression Analysis," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle
Section 2.1

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Financial Reporting and Analysis – Suburban

Claire Munroe, the senior publishing analyst at North Star Securities, has begun to review the recent financial performance of Suburban Publishers, Inc. Suburban, which reports under US GAAP, has a history of purchasing community news groups from around the country and holding them for several years even if they are not initially profitable. The growth of the internet has been difficult on many major newspapers, but community newspapers have been particularly resilient.

At the start of 2010, Suburban purchased 24% of the 1 million outstanding shares of West Reach Community News Group for \$12,000 thousand. West Reach's income and dividends through the end of 2013 are shown in Exhibit 1.

Exhibit 1
West Reach Community News Group Income and Dividends, 2010-2013

(\$ thousands)				
	2010	2011	2012	2013
Net income	2,400	1,800	1,850	2,000
Dividends	500	120	48	418

As Munroe reviewed her working papers, she came across a notation that she had made following the acquisition: "A very strange long-term acquisition for Suburban. West Reach's majority holder, William French (who is now 81 years old), holds 62% of the shares and controls the board with an iron hand. Dividends are paid out according to his needs and preferences. Suburban was unsuccessful in getting any of its preferred candidates elected to the board or exerting any influence on West Reach's dividend policy." Just before Monroe closed her file on this firm, she added, "Nothing has changed since 2010, except, of course, that Mr. French is now a few years older".

At the start of 2011, Suburban purchased 32.5% of the outstanding 2 million shares of Great Lakes Free Press, Inc., for \$1,365 thousand. Great Lakes owned community newspapers in most of the northeastern states. Although the majority of these papers were provided free of charge, they had historically maintained strong revenue streams with their focus on personal interest stories about local individuals and local business advertising primarily from the automotive sector. Exhibit 2 provides details of this acquisition (Part A) and subsequent results (Part B). Great Lakes struggled in 2012 and 2013 with mounting losses and the elimination of its dividend.

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Exhibit 2

Great Lakes Free Press Values at Acquisition and Subsequent Performance, 2011-2013

Part A: Values at Acquisition, 1 January 2011			
(\$ thousands)	Book Value	Fair Value	
Current assets	120	120	
Plant and equipment (P&E)*	2,280	2,964	
Land	<u>1,440</u>	<u>1,476</u>	
	3,840	4,560	
Liabilities	<u>1,200</u>	<u>1,200</u>	
Net assets	2,640	3,360	
*Estimated useful life remaining as of the date of acquisition is 10 years, with straight line depreciation to be used.			
Part B: Performance since Acquisition			
(\$ thousands)	2011	2012	2013
Net Income	1,200	-200	-600
Dividends	504	0	0

Munroe's calculations for Suburban's holdings of Great Lakes at the end of 2013 are summarized in Exhibit 3. She believed that even with government bailouts, there was little chance of a permanent recovery in the automotive sector, and with Great Lakes' heavy reliance on that industry, Suburban would most likely have to treat the investment as being impaired.

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Exhibit 3

**Basis of Munroe's Opinion of Impairment in Great Lakes, as of
year-end 2013**

(\$ thousands)	
Book value of Great Lakes	3,256.00
Fair value of Suburban's investment in Great Lakes	940
Carrying value of Suburban's investment in Great Lakes	1,264.51

At the start of 2013 Suburban decided to provide its publications with a new, fresh look and include more high-quality colored images. To meet this need, Suburban purchased HiQ Printers, which had high speed production printing presses in all of Suburban's distribution areas. Suburban purchased 60% of the company's shares in exchange for its own shares. At the time of the purchase, there were 8 million shares of HiQ Printers outstanding trading at \$14 per share. The fair value of HiQ Printers' net identifiable assets at that time was \$99 million. Exhibit 4 shows the shareholders' equity of both companies prior to the business combination.

Exhibit 4

**Shareholders' Equity for Suburban Publishers and HiQ Printers Prior
to the
Combination in January 2013**

(\$ thousands)	Suburban Publishers	HiQ Printers
Capital stock (no par)	280,000	40,000
Retained earnings	185,000	26,000

After purchasing HiQ Printers, Suburban had excess printing capacity in its older printing facilities. Munroe gathered some information to try and determine the recoverable value of these other facilities, shown in Exhibit 5.

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Exhibit 5
Estimated Recoverable Value of
Suburban's Publishing Assets
(excluding those owned through HiQ Printers)

(\$ thousands)	
Net book value of equipment	120
Expected future cash flows per year	10.8
Undiscounted expected future cash flows	108
Fair value	111
Value in use	61
Estimated remaining life	10 years
Estimated cost of capital	12%

Munroe planned to determine whether these publishing assets were impaired, and if so, what the impact would likely be on Suburban's financial statements.

13. The *most appropriate* way for Suburban to account for changes in the value of its West Reach holdings is to:
- A. include them on the income statement.
 - B. ignore them unless there is an impairment.
 - C. include them in shareholders' equity.

Answer = C

Although Suburban's ownership interest (24%) in West Reach exceeds the amount that is normally deemed sufficient to presume significant influence (20%), the inability to elect directors or influence the firm's policy making indicates a lack of significant influence. It would thus be inappropriate to account for its holdings under the equity method, but given the long-term duration of the planned investment, an available-for-sale investment (or fair value through other comprehensive income) would seem to be the most appropriate. The investment should be remeasured and recognized at fair value on the balance sheet with any unrealized gains or losses arising from fair value reported in equity as other comprehensive income.

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CFA Level II
 "Intercompany Investments," Susan Perry Williams
 Sections 3.3, 5

14. At the end of 2011, the balance in the investment in the associate account for Great Lakes Free Press (in thousands) was *closest* to:
- A. \$1,567.80.
 - B. \$1,591.20.
 - C. \$1,568.97.

Answer = C

Analysis of initial investment (\$ thousands)		
Amount paid in acquisition	1,365	
Book value of acquired equity	<u>858</u>	32.5% × 2,640
Excess paid over book value	507	
Amount attributable to fair value of identifiable assets	<u>234</u>	32.5% × (684 + 36): Excess paid for P&E and Land
Goodwill (residual value)	273	
Calculation of investment in associate account as of year-end 2011 (\$ thousands)		
Start of year investment	1,365.00	Amount paid
+ Share of net income	390.00	32.5% × 1,200
– Share of dividends received	–163.80	32.5% × 504
– Amortization of excess amount paid for PPE	<u>–22.23</u>	32.5% × 684 ÷ 10 years
End of year investment	1,568.97	

CFA Level II
 "Intercompany Investments," Susan Perry Williams
 Sections 5.1, 5.2

15. With regard to Munroe's opinion about the possible impairment of the investment in Great Lakes Free Press, the impairment loss (in thousands) is *closest* to:
- A. \$118.20.
 - B. \$324.51.
 - C. \$425.00.

Answer = B

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Under US GAAP, if the fair value of the investment falls below the carrying amount, and assuming the decline is considered to be permanent (as Munroe expects), the impairment loss is recognized on the income statement and the carrying value of the investment is reduced to its fair value; the impairment loss is the difference between the carrying amount and the fair value (i.e., \$1,264.51 thousand – \$940.00 thousand = **\$324.51** thousand).

CFA Level II

"Intercompany Investments," Susan Perry Williams

Section 5.5

16. Immediately following its business combination with HiQ Printers, the total shareholder's equity (in thousands) on Suburban's consolidated financial statements is *closest* to:
- A. \$532,200.
 - B. \$577,000.
 - C. \$571,800.

Answer = B

The shareholders' equity section of the post-acquisition consolidated balance sheet will consist of the capital stock and retained earnings account of the parent and the non-controlling interest of the minority shareholders. Under US GAAP, the value of the non-controlling interest is equal to the non-controlling interest's proportionate share of the subsidiary's fair value.

A: Amount paid for 60% interest = 60% x \$14/share x 8,000,000 shares = \$67,200 thousand		
Fair value of subsidiary = 8 million shares x \$14/share = \$112,000 thousand		
B: Non-controlling interest = \$112,000 x (1 – 0.60) = \$44,800 thousand		
Shareholders' Equity:	\$ thousands	Calculation
Non-controlling interest	44,800	See B above
Capital stock	347,200	280,000 + 67,200: Original + Issued in acquisition (see A: above)
Retained earnings	<u>185,000</u>	Suburban's retained earnings
Total equity	577,000	

CFA Level II

"Intercompany Investments," Susan Perry Williams

Section 6.4

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17. If Munroe's estimates of recoverability in Exhibit 5 are correct, the impairment loss (in thousands) that will be reported by Suburban following the acquisition of HiQ Printers is *closest* to:
- A. \$12.0
 - B. \$59.0
 - C. \$9.0.

Answer = C

Under US GAAP, first recoverability is assessed by comparing the net book value (NBV) with the undiscounted cash flows: (NBV = 120 > Undiscounted cash flows = 108). Therefore the asset is impaired and the amount of the impairment loss must be determined as:

Impairment loss = Carrying amount – Asset's fair value = 120 – 111 = **9.0**.

CFA Level II

"Long-lived Assets: Implications for Financial Statements and Ratios," Elaine Henry and Elizabeth

A. Gordon

Section 4.1

18. If Munroe's estimates of recoverability in Exhibit 5 are correct, the *most likely* impact on Suburban's financial statements is that:
- A. total asset turnover will increase.
 - B. cash flow from operations will decrease.
 - C. net profit margin will remain unchanged.

Answer = A

Total asset turnover = Sales/Total assets; sales will be unaffected, but assets will be written down, resulting in an increase in the asset turnover ratio.

CFA Level II

"Long-lived Assets: Implications for Financial Statements and Ratios," Elaine Henry and Elizabeth

A. Gordon

Section 4.1

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Financial Reporting and Analysis - Atlantic Preserve

Jim Loris is the Food and Beverage analyst at Eastern Trust & Investments. Jeremy Paul is an intern under Loris's supervision. Loris is planning on reviewing the financial statements of Atlantic Preserves, Inc., in the next few days. The company has recently signed a new collective agreement with its workers, and Loris is interested in seeing how the company's employment costs have been affected. The company prepares its financial statements in accordance with US GAAP, and the new collective agreement became effective 1 January 2014.

Paul extracts portions of the new collective agreement related to the pension plan and mentions to Loris that there have been two changes related to the plan:

- The benefit formula has been changed to $1.75\% \times \text{Final year's salary} \times \text{Number of years of service}$ under the plan. Previously, the same formula was used, but with a factor of 1.65%.
- The vesting period has been changed from four years to three years.

Paul makes the following two comments about these changes to the pension plan:

1. The new formula will have a big impact on income because the past service costs that arise will be expensed immediately.
2. The change to a shorter vesting period will give rise to an actuarial gain.

Loris responds: "The past service costs that arise will be reported in other comprehensive income and amortized on the profit and loss statement over the average service lives of the employees."

Loris provides Paul with the information in Exhibit 1 about John Smith, an employee who has just started working for Atlantic, and other information taken from the company's pension plan disclosures. Loris asks Paul to calculate the pension liability arising from Smith.

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Exhibit 1: Assumptions Relating to the Liability Arising from John Smith's Pension

Pension Plan Details and Assumptions		Employee Details	
Annual wage increase	3.50%	Current salary	\$60,000
Discount rate	7.50%	Date hired	1-Jan-14
Pension Plan Benefit Payments		Expected retirement date	31-Dec-19
Annual payments are paid at year end and continue for the remainder of the retiree's life		Estimated final salary	\$71,261
		Estimated years in retirement	25

Following his calculation of the pension plan liability, Paul asks Loris two questions about the discount rate that is used:

1. Exhibit 1 does not mention how you determined the discount rate that was used. What rate is the most appropriate rate to use?
2. What would be the effect of using a higher discount rate on various components of the company's pension plan obligation?

Loris answers Paul's questions and then provides him with selected information from Note F of the 2013 Annual Report of Atlantic Preserves, shown in Exhibit 2. He tells Paul that he is aware that the company's actual return on pension plan assets exceeds its expected return and asks Paul to use the information in Exhibit 2 to calculate the net periodic pension cost and the total periodic pension cost for Atlantic for 2013.

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Exhibit 2: Selected Information from Note F of Atlantic's 2013 Annual Financial Statements

(\$ thousands)	
Start-of-year pension obligations	72,544
Start-of-year plan assets	60,096
End-of-year pension obligations	74,077
End-of-year plan assets	61,812
Current service cost	1,151
Interest cost	5,441
Actual return on plan assets	5,888
Expected return on plan assets	4,597
Benefits paid to retired employees	5,059
Employer's contributions	887
Amortisation of past service costs	272

19. In regard to Loris and Paul's discussion about the changes in the pension plan arising from the new collective agreement, which comment is *most* accurate?

- A. Paul's first comment about the impact on income
- B. Paul's second comment about the actuarial gain
- C. Loris' response about past service costs

Answer = C

Loris' response about the past service costs is most accurate. Past service costs arise because of the enrichment of the pension benefit to be received under the plan. Under US GAAP, any past service costs will be reported in other comprehensive income and are amortized on the profit

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and loss statement over the average service lives of the employees. Under IFRS, the past service costs are recognized as an expense in the income statement.

CFA Level II

"Employee Compensation: Post-Employment and Share Based," Elaine Henry and Elizabeth A. Gordon

Section 2.1, 2.2, 2.3.2.2

20. At the end of Smith's second year of service, the estimated defined benefit obligation arising from his employment is *closest* to:
- A. \$20,818.
 - B. \$27,802.
 - C. \$20,092.

Answer = A

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1. Determination of annual unit credit (benefit)				
Estimated final salary (Exhibit 1):				\$71,261
Estimated annual (end of year) payment in retirement (six years of service, 2014–2019):		$\$71,261 \times 1.75\% \times 6$		\$7,482.41
Present value of estimated future payments as of the start of retirement (keystrokes using a financial calculator):		PV of 7,482.41 for 25 years at 7.5% ($N = 25, I = 7.5, PMT = 7,482.41$ Mode: End; $PV = ?$)		\$83,406
Annual unit credit at time of retirement per service year:		83,406/6		\$13,901
2. Determination of build-up of pension obligation for the employee				
	Calculation for 2014	2014	Calculation for 2015	2015
Opening obligation		0	From close of 2014	\$9,683
Interest cost at 7.5%		0	$\$9,683 \times 0.075$	\$726
Current service cost Present Value of the unit credit	$\frac{\$13,901}{(1 + 0.075)^5}$	<u>\$9,683</u>	$\frac{\$13,901}{(1 + 0.075)^4}$	<u>\$10,409</u>
Closing obligation		\$9,683		\$20,818

CFA Level II

"Employee Compensation: Post-Employment and Share Based," Elaine Henry and Elizabeth A. Gordon
Section 2.3.3

21. The *best* answer to Paul's first question is to use the:

- A. company's before-tax cost of debt.
- B. company's overall cost of capital.
- C. yield on high quality corporate bonds.

Answer = C

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The yield on high quality corporate bonds is the appropriate discount rate that should be used to calculate the present value of the future benefits because it represents the rate at which the defined benefit obligation could be effectively settled.

CFA Level II

"Employee Compensation: Post-Employment and Share Based," Elaine Henry and Elizabeth A. Gordon

Sections 2.2, 2.3.3, 2.4.1

22. The *least* appropriate answer to Paul's second question is that the:

- A. current service cost would increase.
- B. interest cost may either increase or decrease.
- C. opening obligation would decrease.

Answer = A

The current service cost will decrease, not increase. A higher discount rate means that the present value of the future benefits earned in retirement will be lower and thus the annual unit credit will be lower. Therefore, the current service cost will decrease.

CFA Level II

"Employee Compensation: Post-Employment and Share Based," Elaine Henry and Elizabeth A. Gordon

Section 2.3.2.2

23. The amount of Atlantic Preserve's 2013 periodic pension cost (in \$ thousands) is *closest* to:

- A. 2,267.
- B. 1,995.
- C. 976.

Answer = A

Under US GAAP, the periodic pension cost is calculated as follows:

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	\$ thousands
Current service cost	1,151
Interest cost on the obligation	5,441
Less expected return on plan assets	<u>-4,597</u>
Plus amortization of past service costs	<u>272</u>
Periodic pension cost	2,267

CFA Level II

"Employee Compensation: Post-Employment and Share Based," Elaine Henry and Elizabeth A. Gordon

Section 2.3.2.2, Exhibit 2

24. Atlantic Preserve's total periodic pension cost (in \$ thousands) for 2013 is *closest* to:

- A. 183.
- B. 2,267.
- C. 704.

Answer = C

The total periodic pension cost is the change in the net pension asset or liability excluding the effect of the employer's periodic contribution to the plan.

(\$ thousands)	Pension Obligations	Pension Assets	Net Liability ^a
End of year	74,077	61,812	12,265
Less start of year	<u>72,544</u>	<u>60,096</u>	<u>-12,448</u>
Net change in funded status (decrease in net liability)			183
Employer's contributions (cost)			887
Less decrease in liability			<u>-183</u>
Total periodic pension cost ^b			<u>704</u>
Alternative Calculation:			
Service cost		1,151	
Interest cost		5,441	
Less actual return on plan assets		<u>-5,888</u>	

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Total periodic pension cost	<u>704</u>
^a A net pension liability is a negative funded status. ^b Total periodic pension cost represents the decrease in the net pension liability by \$183 thousand and the employer's contribution of \$887 thousand: $\$887 - \$183 = \$704$ thousand	

CFA Level II

"Employee Compensation: Post-Employment and Share Based," Elaine Henry and Elizabeth A. Gordon

Section 2.4.3

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Equity - Yee

Bryan Yee is a junior analyst at HK Partners, a leading asset manager in Hong Kong. His boss, Brittany Chen, has asked Yee to assist her in analyzing eLeisure, a leading firm in the travel and leisure industry. eLeisure operates an online travel agency in Asia that provides travel products and services to travelers and travel agents. Chen provides Yee with a list of questions to help her finalize her analysis of discount rates as they pertain to the valuation of eLeisure, compare the firm with its industry, and determine intrinsic value estimates for eLeisure's common stock.

Chen first asks Yee to estimate eLeisure's sustainable growth rate, which he does using the information in Exhibits 1 and 2.

Exhibit 1	
Selected eLeisure Income Statement Data	
(HK\$ millions, except shares outstanding)	2014
Sales	3,110.56
Pretax income	551.22
Income taxes	135.48
Net income	415.74
Dividends	103.87
Common shares outstanding (millions)	89.54

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Exhibit 2 Selected eLeisure Balance Sheet Data		
	2014 Cost (HK\$ millions)	2014 Market Value (HK\$ millions)
Cash	490	
Total assets	4,235.58	
Total debt	1,051.96	997
Common shareholders' equity	2,119.41	
Non-controlling interest	<u>580</u>	
Total equity	2,699.41	

The current share price of eLeisure's common equity is HK\$31.28. Chen mentions to Yee that historically, the company has had a ratio of enterprise value (EV) to sales of 1.25x. She asks Yee to use the information in Exhibits 1 and 2 along with this metric to determine whether eLeisure's common shares are appropriately priced.

Chen asks Yee to refine his analysis of the dividend growth rate and discount rates to value eLeisure's equity. Yee looks at eLeisure in more detail and concludes that its expansion potential will likely follow three distinct stages of growth, provided in Exhibit 3. He also determines the long-term return on equity (ROE) for the stock and its required rate of return, which are also presented in Exhibit 3.

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Exhibit 3		
Yee's Estimates for eLeisure		
1. Estimated growth rates for eLeisure's dividends		
2015-2017	2018-2021	Beyond 2021
19%	10%	5%
2. Other estimates for eLeisure		
Long-term ROE		15%
Required rate of return on the stock		11%

With these estimates, Yee determines the intrinsic value of eLeisure common stock using the dividend discount model (DDM).

Chen next instructs Yee to minimize the uncertainty in making assumptions about eLeisure's future earnings and long-term dividend growth by using the residual income model. Yee uses the data in Exhibits 1, 2, and 3 to calculate eLeisure's intrinsic value per common share.

Yee discusses with Chen the best reasons for using the residual income model and provides the following explanation:

"The residual income model's strengths include the fact that it uses readily available accounting data and focuses on economic profitability. Weaknesses include the fact that accounting data can be manipulated by management, the cost of debt capital is assumed to be reflected by interest expense, and terminal values make up a large portion of the value of a firm's equity."

Several firms in the leisure industry in Asia are privately held. Chen asks Yee to provide three key differences between valuing private and public companies. He cites the following differences:

- Private firms are generally smaller than public firms. Being smaller, they can have enhanced growth prospects because of easier access to growth capital.

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- Agency issues are usually greater at private companies.
- Small companies might decide to remain privately held because higher compliance costs may outweigh any other benefits of being public.

25. Using the information in Exhibits 1 and 2, Yee's estimate of eLeisure's sustainable growth rate is *closest to*:

- A. 4.9%.
- B. 14.7%.
- C. 7.4%.

Answer = B

There are two ways to calculate sustainable growth rate (g). The first way is to use the PRAT model (profit margin, retention ratio, asset turnover, and financial leverage, T) (amounts in HK\$ millions):

$$\begin{aligned}
 g &= \text{Retention ratio} \times \text{Profit margin} \times \text{Asset turnover} \times \text{Financial leverage} \\
 &= \frac{(\text{Net income} - \text{Dividends})}{\text{Net income}} \times \frac{\text{Net income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Total assets}} \times \frac{\text{Total assets}}{\text{Common shareholders' equity}} \\
 &= \frac{(415.74 - 103.87)}{415.74} \times \frac{415.74}{3,110.56} \times \frac{3,110.56}{4,235.58} \times \frac{4,235.58}{2,119.41} \\
 &= 0.750 \times 0.134 \times 0.734 \times 2.00 = \mathbf{14.7\%}
 \end{aligned}$$

An alternative approach is to determine g by multiplying the retention ratio by ROE:

$$\begin{aligned}
 g &= \text{Retention ratio} \times \text{ROE} \\
 &= \frac{(\text{Net income} - \text{Dividends})}{\text{Net income}} \times \frac{\text{Net income}}{\text{Common shareholders' equity}} \\
 &= \frac{(415.74 - 103.87)}{415.74} \times \frac{415.74}{2,119.41} \\
 &= 0.750 \times 0.196 = \mathbf{14.7\%}
 \end{aligned}$$

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CFA Level II

"Discounted Dividend Valuation," Jerald E. Pinto, Elaine Henry, Thomas R. Robinson, and John D. Stowe
Section 6.2

26. Using the EV-to-sales ratio approach, Yee discovers that compared with this metric, eLeisure's common shares are *most likely* currently:
- A. properly valued.
 - B. undervalued by 20.7%.
 - C. overvalued by 17.5%.

Answer = A

1. The company's EV/sales multiple is 1.25. With eLeisure's sales of HK\$3,110.56 million, $EV = 1.25 \times HK\$3,110.56 = HK\$3,888.20$ million.
2. eLeisure's enterprise value = Market value (MV) of common equity + MV of debt + Non-controlling interest - Cash and investments $= (HK\$X \times 89.54 \text{ million}) + HK\$997.00 \text{ million} + HK\$580.00 \text{ million} - HK\$490.00 \text{ million} = HK\$3,888.20 \text{ million}.$ Solving for $X = HK\$31.28$.
3. With a current share price of HK\$31.28, eLeisure's shares are currently properly valued according to this metric.

CFA Level II

"Market-Based Valuation: Price and Enterprise Value Multiples," Jerald E. Pinto, Elaine Henry, Thomas R. Robinson, and John D. Stowe
Section 4.3

27. Based on Yee's growth estimates and the information in Exhibits 1 and 3, the amount that the terminal value component of its intrinsic value contributes to eLeisure's stock price at the end of 2014 is *closest* to:
- A. HK\$21.73
 - B. HK\$20.30.
 - C. HK\$24.12.

Answer = C

Using the DDM approach and Yee's three-stage growth estimates, the contribution that the terminal value provides to eLeisure's stock price at the end of 2014 is calculated as follows:

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1. Determine the current dividend by taking the 2014 dividend of HK\$103.87 million divided by the shares outstanding of 89.54 million = HK\$103.87 million/89.54 million shares = HK\$1.16/share.

2. Then determine the value of the dividend when it enters its terminal growth stage.

Dividend Behavior during the Three Growth Periods		
Period of growth	Formula	Value
First	$1.16 \times (1.19)^3$	1.954
Second	$1.954 \times (1.10)^4$	2.862
Third	All future growth at 5%	
3. The present value of all future dividends as of 2021 can be determined using the growing perpetuity formula.		
$V_T = \frac{D_{T+1}}{r - g} = \frac{D_T \times (1 + g)}{r - g}$	$V_{2021} = \frac{2.862 \times (1 + 0.05)}{0.11 - 0.05} = \text{HK\$}50.08$	
4. The present value as of 2014 of this terminal value is:	$V_{2014} = \frac{50.08}{(1 + 0.11)^7} = \text{HK\$}24.12$	

CFA Level II

"Discounted Dividend Valuation," Jerald E. Pinto, Elaine Henry, Thomas R. Robinson, and John D. Stowe
Section 5.4

28. Using the residual income model and Exhibits 1, 2, and 3, Yee's estimate of eLeisure's intrinsic value per share is *closest* to:

- A. HK\$39.45.
- B. HK\$32.27.
- C. HK\$23.67.

Answer = A

The residual income model is:

$$V_n = B_0 + \frac{ROE - r}{r - g} B_0$$

where

B_0 = Book value per common share at end of 2014 (beginning of 2015)

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ROE = Long-term return on equity

r = Cost of equity

g = Long-term dividend growth rate

The book value per common share is calculated by dividing the common shareholders' equity in Exhibit 2 by the common shares outstanding in Exhibit 1:

Common shareholders' equity	HK\$2,119.41 million
÷ Common shares outstanding	<u>89.54 million</u>
= Book value per common share	HK\$23.67

The intrinsic value for eLeisure's common shares is:

$$V_n = \text{HK\$}23.67 + \frac{0.15 - 0.11}{0.11 - 0.05} 23.67$$

CFA Level II

"Residual Income Valuation," Jerald E. Pinto, Elaine Henry, Thomas R. Robinson, and John D. Stowe
Sections 3.2, 3.3

29. Which of Yee's explanations of the strengths and weaknesses of the residual income model is *least* accurate?
- A. The explanation about accounting data
 - B. The explanation about debt capital
 - C. The explanation about terminal values

Answer = C

Terminal values do not make up a large portion of the total present value of a firm's equity. Current book value often captures a large portion of total value. Therefore, terminal value may not be a large component of total value.

CFA Level II

"Residual Income Valuation," Jerald E. Pinto, Elaine Henry, Thomas R. Robinson, and John D. Stowe
Section 4.1

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30. Which of the differences cited by Yee about private and public companies is *most* accurate?

- A. The differences in compliance costs
- B. The differences in enhanced growth prospects
- C. The differences in agency issues

Answer = A

For small companies, the cost of operating as a public company and its related compliance costs may outweigh enhanced access to capital that comes with being a public company.

CFA Level II

"Private Company Valuation," Raymond D. Rath

Section 2.1.1

"Dividends and Share Repurchases: Analysis," Gregory Noronha and George H. Troughton

Section 2.4.3

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Equity – GreenSnacks

Peter Tanner recently accepted a position as a domestic equity analyst with a large US pension fund. The fund uses a bottom-up team approach to stock selection. The fund's equity manager, Cindy Bradley, is responsible for the results of the domestic equity portfolio.

Tanner's first assignment is to evaluate a packaged foods company, GreenSnacks, Inc. (GNSK), which trades on the NASDAQ at a current price of \$21.875 per share. His analysis is to include a long-term outlook for the company in the context of the well-established packaged foods industry, which is dominated by several large companies in the United States.

The major players compete vigorously for market share. The industry has been growing at a rate very similar to that of GDP for many years. GNSK competes in the rapidly growing health food category, which has been gaining about 1%-2% of relative share per year within the broader packaged foods industry because of external factors, such as changes in social preference and an aging population.

Originally a spin-off from one of the industry's key players, GNSK has shown promising growth for the past few years because of a new and bold process it developed to enhance the preservation, packaging, and distribution of fruity snacks. GNSK has obtained several important patents for this process, and the end result is more healthy, moist, and flavorful snacks. GNSK's innovative advancement took time and care to develop, and management was willing to stay the course to develop its new process. The combination of its efforts has allowed GNSK to create products that maintain a fresh taste without preservatives and with a shelf-life much longer than the established products of the leading brands.

Although the healthy snack category has been gaining market share rapidly because of social changes, GNSK was not able to break through the established shelf-space barrier controlled by the large competitors until its new process was perfected. As national chains begin to pick up the product line, GNSK is experiencing substantial market share gains and accelerating sales. The outlook for the company's future sales growth exceeds 17%, and profit margins are increasing well beyond the levels of competitors.

Tanner expects short-term rapid earnings growth of 20% in 2014 for GNSK, with the rate of growth linearly diminishing over the next five years to match industry conditions thereafter. He assumes that starting in Year 6, GNSK's long-term dividend growth rate will be equal to the current level of sustainable growth rate for the industry. Given these assumptions and the data in Exhibit 1, Tanner decides to use the H-model for valuing GNSK's stock.

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Exhibit 1: Selected Financial Information for the Fiscal Year Ending 31 December 2013

	GNSK	Industry Average
Return on equity (%)	23.1	12.8
Earnings per share (EPS) (\$)	2.45	n/a
Dividend payout ratio (%)	25	65
Required return (%)	n/a	11
Trailing dividend yield (%)	2.8	3.7

Note: n/a not available

Tanner meets with Bradley for her advice. Bradley states that because the packaged foods industry is mature and stable, she would prefer that Tanner calculate the implied long-term dividend growth rate for GNSK using the Gordon growth model. Furthermore, Bradley believes that the required return and dividend yield for the industry are the most stable indicators and should be used in the valuation computations.

Bradley suggests that Tanner also analyze the investment appeal of industry peer Star Cakes (STCK). She provides the following data and assumptions for STCK:

- The company paid a dividend of \$2.48 last year.
- Its earnings are dropping about 2% every year permanently.
- I assign a required return of 7.4% for this company because of its low beta.

After his meeting with Bradley, Tanner discusses different valuation methods with his colleagues, Marcia Stephens, Dale Mathews, and Kevin Baldrige. His colleagues make the following statements:

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Stephens: Free cash flow valuation is especially appropriate for investors who want to take a control perspective in takeovers. Also, free cash flow to equity is the cash flow available to be distributed to shareholders without impairing the company's value.

Mathews: Remember that the Gordon growth model is based on indefinitely extending future dividends and the intrinsic value derived by the model is very sensitive to small changes in the assumed growth rate and required rate of return.

Baldrige: You can use the residual income approach as well; it is a simpler model that does not require holding of the clean surplus relationship, and the valuation is not impacted by book values.

Tanner prepares a list of issues he needs to consider and begins his analysis for his report.

31. GNSK can *best* be described as being in which of the following growth stages?

- A. Transition
- B. Growth
- C. Mature

Answer = B

GNSK is in the growth stage because it is expanding rapidly and enjoying the benefits of the health food market, which is also growing rapidly. GNSK is also experiencing high and growing profit margins as well as abnormally high earnings per share growth, which are all indicative of a company in its growth phase.

CFA Level II

"Discounted Dividend Valuation," by Jerald Pinto, Elaine Henry, Thomas Robinson, and John Stowe
Section 4.7

32. Which of the following strategies did GNSK *most likely* pursue to achieve its success?

- A. Adaptive
- B. Shaping
- C. Visionary

Answer = C

GNSK's strategy is best described as visionary. The slow growth of the packaged food industry required a bold strategy that GNSK undertook to improve its own growth profile. This approach

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is characteristic of a visionary strategy. The process to develop its healthier foods with longer shelf lives took time and care, and management was willing to stay the course to develop its new process. Both of these decisions are also characteristic of the visionary strategy.

CFA Level II

“Your Strategy Needs a Strategy,” Martin Reeves, Claire Love, and Philipp Tillmanns

33. According to the valuation approach that Tanner decides to use and data from Exhibit 1, the expected rate of return for GNSK is *closest* to:
- A. 9.6%.
 - B. 8.5%.
 - C. 12.2%.

Answer = B

The H-model that Tanner decides to use is a variant of the two-stage dividend discount model. It assumes growth begins at a high rate and declines linearly throughout the super-normal growth period until it reaches a normal rate at the end. In the case of GNSK, the H-model is appropriate for estimating the required return because Tanner expects extraordinary earnings growth of 20% next year with the rate of growth diminishing over time to match industry conditions in Year 6.

$$r = \left(\frac{D_0}{P_0} \right) [(1 + g_L) + H(g_s - g_L)] + g_L$$

H = Half-life in years of the super-normal growth rate = $5 \times 0.5 = 2.5$

$D_0 = 2.45 \times 0.25 = \0.6125

g_L = Sustainable growth rate for the industry = $\text{ROE} \times (1 - \text{payout}) = 0.128 \times (1 - 0.65) = 0.0448$ or 4.48%

g_s = Short-term growth rate of GNSK = 0.20

$$r = \left(\frac{.6125}{21.875} \right) \{ (1 + 0.0448) + 2.5(0.20 - 0.0448) \} + 0.0448$$

$$= (0.028) \{ (1.0448) + 0.388 \} + 0.0448$$

$$= 0.08492 \text{ or } 8.5\%$$

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CFA Level II

"Discounted Dividend Valuation," Jerald Pinto, Elaine Henry, Thomas Robinson, and John Stowe
Sections 5.2, 5.6

34. Using Bradley's assumptions regarding GNSK and the data from Exhibit 1, GNSK's implied long-term dividend growth rate is *closest* to:
- A. 8.0%.
 - B. 7.0%.
 - C. 7.3%.

Answer = B

$$V_0 = \frac{D_0(1+g)}{r-g}$$

Rearranging:

$$\frac{D_0}{P_0} = \frac{r-g}{1+g}$$

Let $D_0/P_0 = d$ and rewrite the equation:

$$g = \frac{r-g}{1+d}$$

Using the industry data in Exhibit 1:

$$g = \frac{0.11 - 0.037}{1 + 0.037} = 0.0704 \text{ or } 7.0 \%$$

Alternatively:

$$0.037 = (0.11 - g)/(1 + g)$$

$$0.037(1 + g) + g = 0.11$$

$$0.037 + 0.037g + g = 0.11$$

$$1.037g = 0.11 - 0.037$$

$$g = 0.073 / 1.037 = 0.0704 \approx 7.0\%$$

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CFA Level II

"Discounted Dividend Valuation," Jerald Pinto, Elaine Henry, Thomas Robinson, and John Stowe
Section 4.1

"Market-Based Valuation: Price and Enterprise Value Multiples," Jerald Pinto, Elaine Henry,
Thomas Robinson, and John Stowe
Section 3.5.2

35. Given Bradley's suggestions and assumptions, STCK's intrinsic value is *closest* to:

- A. \$26.38.
- B. \$32.84
- C. \$25.85.

Answer = C

$V_0 = D_1 / (r - g)$ where $D_1 = D_0 (1 + g)$.

$$V_0 = \frac{2.48(1 + .02)}{[0.074 - (-0.02)]}$$

$$= 2.43 / 0.094 = \mathbf{\$25.85}$$

CFA Level II

"Discounted Dividend Valuation," by Jerald Pinto, Elaine Henry, Thomas Robinson, and John Stowe
Section 4.1

36. Which valuation methodology statement made by Tanner's colleagues is *least* accurate? The statement made by:

- A. Mathews.
- B. Baldridge.
- C. Stephens.

Answer = B

Baldridge's statement is least accurate. The residual income approach uses the book value of equity, and it requires that the clean surplus relationship holds.

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CFA Level II

"Discounted Dividend Valuation," Jerald Pinto, Elaine Henry, Thomas Robinson, and John Stowe
Section 2.2

"Residual Income Valuation," Jerald Pinto, Elaine Henry, Thomas Robinson, and John Stowe
Section 3.1

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Fixed Income - Attersee

Jacob Bauer is a credit analyst for Attersee Partners, an investment management firm. Attersee was recently hired by Linz Corporation to manage their pension fund portfolio. Bauer is meeting with Isabella Huber, the senior benefits manager at Linz, to discuss the management of the portfolio's holdings, which are primarily bonds.

During the meeting, Huber asks Bauer how bonds will be selected for the portfolio. Bauer explains that bonds will be selected based on a number of characteristics including duration, industry sector, and liquidity, but that credit risk is the primary characteristic used to identify underpriced bonds. He continues, "We measure the credit risk of a bond by calculating its expected loss. This is equal to the product of the bond's probability of default and its loss given default. The expected loss is compared to the difference between the price of the bond and the price of an otherwise identical credit risk free bond to determine if the bond is fairly priced. We only purchase bonds that we believe are substantially underpriced."

Huber says she is more familiar with other measures of credit risk, specifically credit ratings, but is not sure exactly how they work. Bauer explains some of their characteristics with three statements:

Statement 1: A borrower's credit rating is a summary of a complex analysis of its credit history.

Statement 2: Credit rating systems provide ordinal rather than cardinal rankings of borrowers' credit worthiness.

Statement 3: The default probabilities of debts from different borrowers that have the same credit rating will often be different.

Huber adds that she has read about two weaknesses of credit ratings. First, they tend to fluctuate over time in response to changes in the business cycle. Second, the compensation structure of third party credit-rating agencies creates potential conflicts of interest.

Huber also mentions an article she read about using structural models of credit risk as an alternative to traditional measures like credit ratings. The article explained that owning all of a firm's risky debt is equivalent to owning a riskless bond with the same face value and maturity and taking a position in an option on the issuer's assets. Bauer agrees and identifies the option position as short a European put option on the assets with a strike price equal to the face value of debt and maturity equal to the maturity of the debt. [Q4] Huber asks Bauer whether he uses the structural model to measure credit risk. Bauer replies, "There are numerous problems with estimating the structural model." He provides three examples.

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Problem 1: The structural model assumes that interest rates are constant over time.

Problem 2: The structural model requires estimating changes in asset return volatility related to the business cycle.

Problem 3: The structural model relies on accounting statement data, rather than market prices, and is therefore subject to being manipulated by the firm.

Expanding on his previous statements, Bauer states that the structural model also requires that a firm's assets be publicly traded, which is rarely the case. Because this creates a substantial problem with estimating the structural model, a reduced form model can be used instead. However, reduced form models are based on a number of different assumptions that must be true in order for the model to be valid. These include the assumptions that the borrower have issued a zero coupon bond that currently trades and that the riskless rate of interest is fixed over the life of the debt being analyzed. Further, reduced form models assume that, given a default, the recovery rate is independent of the state of the economy.

37. Which of the following adjustments to the credit risk measure Bauer currently uses to select mispriced bonds for the portfolio is *least likely* to improve his ability to correctly identify a mispricing?
- A. Making adjustments for risk-neutral probabilities of default
 - B. Making adjustments for the present value of future cash flows
 - C. Making adjustments for the expected recovery rate

Answer = C

The expected loss is the product of the probability of default and the loss given default. The loss given default already reflects the expected recovery rate, so no further adjustment is needed for that.

CFA Level II

"Credit Analysis Models," Robert A. Jarrow and Donald R. van Deventer

Section 2

38. Which of Bauer's statements regarding credit ratings is *least likely* correct?
- A. Statement 3
 - B. Statement 1
 - C. Statement 2

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Answer = A

Debts of different borrowers that have the same credit rating will have roughly the same probabilities of default.

CFA Level II

"Credit Analysis Models," Robert A. Jarrow and Donald R. van Deventer
Section 3

39. Is Huber *most likely* correct about the weaknesses of credit ratings she identifies?

- A. Yes
- B. No, she is incorrect about potential conflicts of interest
- C. No, she is incorrect about the impact of the business cycle

Answer = C

A weakness of credit ratings is that they tend to remain stable over the business cycle even though the probability of default changes depending on the state of the economy.

CFA Level II

"Credit Analysis Models," Robert A. Jarrow and Donald R. van Deventer
Section 3

40. Is Bauer *most likely* correct regarding the option position used in the credit risk model?

- A. Yes
- B. No, it should be an American option
- C. No, its strike price should be the present value of the face value of the debt

Answer = A

Owning all of a corporation's debt with face value F and maturity T is equivalent to owning a riskless bond with face value F and maturity T and selling a European put option on the corporation's assets with strike price F and maturity T .

CFA Level II

Credit Analysis Models," Robert A. Jarrow and Donald R. van Deventer
Section 4.1

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41. Which of the problems Bauer describes regarding estimation of the structural model of credit risk is *most likely* correct?

- A. Problem 2
- B. Problem 3
- C. Problem 1

Answer = C

The structural model assumes that interest rates are constant over time, which is not true in practice and a particular problem for a model of debt value because debt is subject to substantial interest rate risk.

CFA Level II

"Credit Analysis Models," Robert A. Jarrow and Donald R. van Deventer
Section 4.4

42. Which of the assumptions of reduced-form models that Bauer describes is *most likely* a true assumption of these models?

- A. The assumption regarding riskless rate of interest
- B. The assumption regarding recovery rate
- C. The assumption regarding the borrower's existing bonds

Answer = C

A reduced-form model requires that the borrower have a zero-coupon bond that trades in arbitrage-free and frictionless markets.

CFA Level II

"Credit Analysis Models," Robert A. Jarrow and Donald R. van Deventer
Section 4.4

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Derivatives – Merimar

Tyra Merinar is a portfolio manager at Ridge Row Capital Advisors (RRCA), a hedge fund based in Charlottesville, Virginia. Merinar is meeting with two assistant portfolio managers, Vinay Jani and Zhong Geng, to review the performance of investments made by RRCA and to evaluate potential new investments. At this meeting, they will discuss two recent investments, an equity swap and a swaption, as well as two potential new investments.

RRCA entered into a one-year equity swap 30 days ago. Under the terms of the swap, the fund will receive the return on the S&P/ASX 300 Metals & Mining Index and pay a fixed annual interest rate of 4.8% on notional principal of \$75,000,000. The swap calls for quarterly payments. At the time the swap was initiated, 30 days ago, the value of the S&P/ASX 300 was 3,250. The value of the S&P/ASX 300 today is 3,738. Merinar wants to determine the market value of the equity swap today using the current term structure of interest rates presented in Exhibit 1.

Exhibit 1: Term Structure of Interest Rates (Equity Swap)

Days	Libor (%)
60	1.42
150	1.84
240	2.12
330	3.42

Three months ago, RRCA purchased a European receiver swaption that is exercisable into a two-year swap with semiannual payments. The swaption has a semiannual exercise rate of 2.75% and a notional principal of \$25,000,000. The swaption has just expired and Merinar asks Jani to determine its cash settlement using the term structure presented in Exhibit 2.

Exhibit 2: Term Structure of Interest Rates (Swaption)

Days	Libor (%)
180	1.95
360	3.68
540	4.11
720	4.65

The meeting's focus turns to potential new investments. Geng has been studying an investment strategy

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that involves potential changes in credit ratings of individual securities. Geng states,

I have been evaluating bonds of Onex Corporation, which are currently rated BBB. Onex has just announced an acquisition that we believe will weaken its credit metrics over the next two years. But longer term, say four to five years, Onex should generate enough cash flow to improve credit quality to pre-acquisition levels. We could use credit default swaps (CDS) on Onex Corporation to take advantage of this improvement. The best way to do this is to buy CDS on Onex Corporation expiring in five years and sell CDS on Onex expiring in two years.

Merinar asks Jani to assess potential mispricing in equity futures markets with a view to implementing an investment strategy to take advantage of any mispricing. Specifically, she asks him to evaluate a futures contract on the S&P MidCap 400 Index that expires in 145 days. The annual risk-free rate is 3.5%, and the index is at 840 today. The accumulated value of dividends reinvested over the life of the futures contract is expected to be \$3.15 per contract.

Merinar explains the investment strategy to be implemented if the stock index futures contract is mispriced. She states, "If futures sell for less than our fair value calculation, the appropriate strategy would be to purchase futures on the index and short the index."

Merinar closes the meeting by asking if Geng and Jani can explain the relationship between futures prices and expected spot prices. Geng responds, "Futures prices are an accurate estimate of expected future spot prices." Jani argues, "I disagree. Expected spot prices are equal to futures prices plus a risk premium." Merinar concludes the discussion saying, "You are both incorrect. Expected spot prices are equal to futures prices minus a risk premium."

43. Using the information provided in Exhibit 1, the market value of the equity swap is *closest* to:

- A. \$9,997,500.
- B. \$7,717,500.
- C. \$7,665,000.

Answer = A

Per \$1 of notional principal, the market value of the equity swap is calculated as follows:

$$\left(\frac{3.738}{3.250} \right) - 0.9696 - (0.012)(0.9976 + 0.9924 + 0.9861 + 0.9696) = 0.1333$$

The market value of the swap = $0.1333 \times \$75,000,000 = \$9,997,500$

The appropriate present value factors are provided in the following table:

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$B_{30}(90)$	0.9976
$B_{30}(180)$	0.9924
$B_{30}(270)$	0.9861
$B_{30}(360)$	0.9696

For example, $B_{30}(90)$ is calculated as

$$\frac{1}{1 + \left[0.0142 \times \left(\frac{60}{360} \right) \right]} = 0.9976$$

CFA Level 2

"Swap Markets and Contracts," Don M. Chance

Section 4.2.3

44. Using the information in Exhibit 2, the market value of the receiver swaption is *closest* to:

- A. \$687,500.
- B. \$495,508.
- C. \$106,250.

Answer = B

The market value of the receiver swaption is calculated as follows:

$$\text{Max}[0, (0.0275 - 0.0223)] \times (0.9903 + 0.9645 + 0.9419 + 0.9149) \times \$25,000,000 = \$495,508$$

The appropriate present value factors are provided in the following table:

$B_0(180)$	0.9903
$B_0(360)$	0.9645
$B_0(540)$	0.9419
$B_0(720)$	0.9149

For example, $B_0(180)$ is calculated as

$$\frac{1}{1 + 0.0195 \times \left(\frac{180}{360} \right)} = 0.9903$$

Other present value factors are calculated in a similar manner.

The fixed rate is calculated as follows:

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$$\frac{1.0 - B_0(h_n)}{\sum_{j=1}^n B_0(h_j)},$$

$$\frac{1.0 - 0.9149}{0.9903 + 0.9645 + 0.9419 + 0.9149} = 0.0223$$

The annualized rate = $0.0223 \times 2 = 0.0446$.

CFA Level 2

"Swap Markets and Contracts," Don M. Chance

Section 4.2.1 and 6.4

45. Is Geng's strategy to take advantage of his credit expectations *most likely* appropriate?
- A. No; the appropriate strategy would be to buy two-year CDS and sell five-year CDS for Onex Corporation
 - B. Yes
 - C. No; the appropriate strategy would be to sell two-year CDS and buy five-year CDS for Onex Corporation

Answer = A

Since Geng expects credit ratings for Onex Corporation bonds to weaken over the near term up to two years, and then strengthen over the longer term (five years), the appropriate strategy is to buy two-year CDS and sell five-year CDS. The two-year CDS would provide a hedge against short-term volatility, and the sale of the five-year CDS would partially fund the purchase of two-year CDS. This trade is a curve flattening trade.

CFA Level 2

"Credit Default Swaps," Brian Rose and Don M. Chance

Section 4.1

46. Assuming a 365-day year, the S&P MidCap 400 Index futures price is *closest* to:
- A. 848.11.
 - B. 854.71.
 - C. 836.85.

Answer = A

The S&P MidCap 400 futures price is $840 \times (1.035)^{(145/365)} - 3.15 = \mathbf{848.41}$.

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CFA Level 2
"Futures Markets and Contracts," Don M. Chance
Section 7.3

47. Is Merinar's investment strategy using stock index futures contracts *most likely* correct?
- A. No; the correct strategy would be to only purchase stock index futures
 - B. Yes
 - C. No; the correct strategy would be to purchase the stock index and sell stock index futures

Answer = B

Merinar is correct. If stock index futures are underpriced, the correct arbitrage strategy would be to short the stock index and purchase stock index futures. This approach will effectively ensure that Merinar has borrowed money at the risk-free rate and repaid at a rate less than the risk-free rate.

CFA Level 2
"Futures Markets and Contracts," Don M. Chance
Section 7.3

48. Who correctly states the relationship between futures prices and expected spot prices?
- A. Geng
 - B. Jani
 - C. Merinar

Answer = B

Jani is correct. The futures price is equal to the expected spot price minus a risk premium. Alternatively, the expected spot price equals the futures price plus a risk premium.

CFA Level 2
"Futures Markets and Contracts," Don M. Chance
Section 7.1.9

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Alternative Investments - Permian

Margaret White is the chief investment officer at Permian Basin Associates (PBA), a commodities trader and fund manager. She asks Thomas Davidson, an analyst, to research energy prices, which have been falling. Davidson collects the oil spot and futures prices as of 31 October 2014, which are presented in Exhibit 1.

Exhibit 1 Brent Crude Oil Spot and Futures Prices as of 31 October 2014	
	Price per Barrel
Spot price	USD85.64
Feb-15	USD80.36
May-15	USD80.23
Aug-15	USD80.12

White and Davidson meet to prepare for PBA's board of directors meeting. White has been asked by the board to address issues dealing with an expansion of the firm's product offering, discuss performance evaluation of its existing funds, and offer guidance about how the return expectations for the fund are determined.

Just prior to the meeting, Saudi Arabia announces that it will increase its crude oil production from 9.6 million barrels per day to 9.9 million barrels per day.

PBA wants to launch a precious metals fund to expand the firm's products. In Exhibit 2, Davidson comments on the characteristics of three common investment approaches that the firm might adopt.

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Exhibit 2 Alternative Investment Approaches for the Proposed Precious Metals Fund	
Investment in	Davidson's Comment
The underlying metals	This investment will involve storage costs and create an administrative burden for PBA.
A portfolio of precious metals futures	This investment will enable PBA to profit from price movements without dealing with the logistics of direct purchase.
A portfolio of precious metals companies	This investment will produce returns that strongly correlate with changes in the underlying commodity prices.

White believes that PBA should evaluate the performance of its funds by comparing each with its underlying subindex and provides Exhibit 3, which shows the return components of the Goldman Sachs Commodities Index(GSCI).

Exhibit 3 Return Components of the Goldman Sachs Subindexes, 1970-2006						
	Spot Return		Roll Return		Collateral Return	
	μ	σ	μ	σ	μ	σ
Agricultural	4.60%	19.68%	-3.86%	5.60%	6.15%	0.87%
Energy	7.87	31.14	2.55	7.64	5.26	2.03
Industrial metals	7.52	22.62	-1.07	6.31	6.21	0.93
Livestock	4.02	19.41	1.2	8.26	6.17	0.95
Precious metals	8.96	23.13	-6.22	2.49	6.24	0.91

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Davidson comments on the fact that the roll returns in Exhibit 3 for the agricultural and energy sectors are quite different. He says, "My understanding is that the

- roll returns of both indexes reflect the respective storage costs associated with each class of commodity,
- roll return of the agricultural subindex is negative when the agricultural futures market has been in contango, and
- roll return of the energy subindex is positive when energy futures prices have been lower than spot prices."

White says, "At the board meeting, we will be asked how we arrived at our commodity futures return expectations. How should we respond?"

Davidson replies, "There are three useful models of expected return. These are the

- insurance perspective, in which hedgers hold commodity inventories and seek to mitigate price risk by buying commodity futures;
- hedging pressure hypothesis, in which investors will receive a risk premium that is a positive excess return for going short in a 'normal contangoed' commodity futures market; and
- theory of storage, which predicts a direct relationship between the level of inventories and the convenience yield."

49. Based on Exhibit 1, the forward curve of oil from February 2015 through August 2015 can *best* be described as being:

- A. flat.
- B. in contango.
- C. in backwardation.

Answer = C

The spot price of oil is USD85.64 and the August 2015 price is USD80.12, so the term structure curve is negative. When the futures price is lower than the spot price, the forward curve is said to be in backwardation.

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"A Primer on Commodity Investing," Frank J. Fabozzi, Roland Füss, and Dieter G. Kaiser
Section: Prices at the Commodity Futures Exchanges

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50. Based on the news announcement about Saudi Arabia, the convenience yield is *most likely* to:

- A. decrease.
- B. remain unchanged.
- C. increase.

Answer = A

Convenience yield is the monetary benefit from holding a commodity physically instead of being long the respective futures and is affected in large part by inventory levels. Supply surpluses lower the spot price and the convenience yield.

CFA Level II

"A Primer on Commodity Investing," Frank J. Fabozzi, Roland Füss, and Dieter G. Kaiser
Section: Commodity Sectors

51. Which of Davidson's statements regarding the alternative investment approaches for the precious metal fund is *least* appropriate?

- A. The strategy involving a portfolio of precious metals futures
- B. The strategy involving direct investment in the underlying metals
- C. The strategy involving a portfolio of precious metals companies

Answer = C

The returns from a portfolio of precious metals companies are only slightly correlated with commodity returns. These publicly traded natural resource companies reflect other price-relevant factors, such as the strategic position of the company, management quality, capital structure, expectations and ratings of revenue and profit growth, risk sensitivity, information transparency, and information credibility. There may also be temporary market disequilibrium, especially for stocks with low free float, in which even a small number of transactions can cause major reactions in the stock price that have nothing to do with the underlying commodity fundamentals. In addition, natural resource companies are subject to operational risk caused by human or technical failure, internal regulations, or external events.

CFA Level II

"A Primer on Commodity Investing," Frank J. Fabozzi, Roland Füss, and Dieter G. Kaiser
Section: Commodities as an Asset Class of Their Own

52. Based on Exhibit 3, the excess return for the livestock subindex is *closest* to:

- A. 5.2%.

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- B. 2.8%
- C. 11.4%.

Answer = A

The excess return is the sum of the spot return and the roll return.

Excess return = Spot return + Roll return = 4.02 + 1.20 = **5.22%**.

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"A Primer on Commodity Investing," Frank J. Fabozzi, Roland Füss, and Dieter G. Kaiser
Section: Prices at the Commodity Futures Exchanges

53. Which of Davidson's comments regarding roll return is *most likely* accurate?

- A. His statement regarding storage costs
- B. His statement regarding the agricultural subindex
- C. His statement regarding the energy subindex

Answer = C

When the futures prices are lower (higher) than spot prices, the futures market is said to be in backwardation (contango), which results in a positive (negative) roll return. There are storage costs associated with both agriculture and energy, so a difference in storage costs would not explain the difference in the roll return.

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Section: Prices at the Commodity Futures Exchanges

54. Which of Davidson's descriptions of the futures return models is *most likely* correct?

- A. The description of the theory of storage
- B. The description of the hedging pressure hypothesis
- C. The description of the insurance perspective

Answer = B

The hedging pressure hypothesis states that investors will receive a risk premium that is a positive excess return for going short in a "normal contangoed" commodity futures market. The insurance perspective assumes that hedgers hold long positions in the underlying commodity

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and short positions in the futures to hedge their risk. The theory of storage predicts an inverse relationship between the level of inventories and the convenience yield.

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Section: Models of Expected Returns

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Portfolio Management - TRS

Elizabeth Robbin is the new chief investment officer at Teacher Retirement Systems (TRS), which oversees 21 funds from 11 different firms. She plans to implement new performance measurement tools for selecting and evaluating TRS's managers.

Robbin is meeting with her staff to gauge what they know about manager selection and evaluation. She starts by asking her staff to share with her their understanding of the term "value added." She notes the following responses from three analysts, David Gladden, Agnes Wert, and Sandra Marano.

Gladden: A manager adds value when the portfolio return is greater than that of its benchmark.

Wert: A manager adds value when he produces a positive rate of return.

Marano: Value added can come from multiple sources, including asset allocation and security selection.

Robbin reviews data compiled by Gladden for Bosphorus Investment Advisers for the meeting. After reviewing Gladden's work, Robbin requests that the analysts include the information ratio in all future exhibits.

Exhibit 1
Bosphorus Investment Advisers: Selected Statistics

Fund average annual return (%)	18.54	Benchmark standard deviation	9.55
Fund standard deviation	10.7	Sharpe ratio	1.59
Benchmark average annual return (%)	18.14	Active risk	1.56

Because the analysts are unfamiliar with the use of the information ratio, Robbin explains how it might be useful in investment manager selection and in choosing the level of active portfolio risk. She asks each analyst to make an observation about his or her understanding of the information ratio.

Marano: The information ratio will change as the active weights deviate from the benchmark weights.

Gladden: Because TRS's investment policy prohibits short positions, TRS would be unable to take advantage of any optimized portfolios with increased active risk.

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Wert: The information ratio appears to be the best criterion to evaluate the past performance of our active managers.

Robbin points out that the Sharpe ratio and the information ratio are both useful tools in evaluating portfolio managers and asks Gladden to explain some of the important differences between the two.

Gladden notes, "The information ratio is a measure of relative expected or realized reward to risk whereas the Sharpe ratio measures the absolute risk-return trade-off of a portfolio. Sharpe ratios help investors focus on the relative value added by active management. Although the information ratio is not affected by the addition of cash or leverage, the Sharpe ratio is affected by the addition of either."

Robbin then introduces the Fundamental Law of Active Management to her analysts, illustrating it with a graphic called the "correlation triangle." "This graph explains how a manager's forecasted returns, decisions about the portfolio's active weights, and realized active returns are related to each other," she says.

Wert observes, "That makes sense. It is difficult to add value if the manager's forecasts do not correspond at least somewhat to the realized active returns. Also, if the portfolio manager does not overweight securities for which he has forecasted the best relative returns, he will not generate positive relative returns."

Lastly, Robbin illustrates to her team how they might apply the Fundamental Law of Active Management in evaluating the performance of Kariba Investment Management.

Robbin observes, "Kariba may be overstating its expected active return. Because Kariba rebalances weekly, it claims that its number of independent decisions is high. However, some of these securities (exchange-traded funds) may cluster in economic regions where the same general analysis applies to several securities. That would mean that Kariba's breadth is in fact much lower than stated. Furthermore, Kariba asserts that each security is independently evaluated. That may not be true either. For example, a strategy that favors a particular economic region will likely persist for several months, and therefore, the investment decisions are not independent. Again, the result would be a lower breadth."

55. Which statement regarding the measurement of value added is *least likely* to be correct?

- A. The statement made by Gladden
- B. The statement made by Wert
- C. The statement made by Marano

Answer = B

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Positive return alone does not reflect value added. The value added of an actively managed portfolio (R_A) is the difference between the return on the portfolio (R_P) and the return on the benchmark portfolio (R_B).

$$R_A = R_P - R_B$$

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Section 2.2

56. Based on the information presented in Exhibit 1, the information ratio for Bosphorus is *closest* to:
- A. 0.53.
 - B. 0.26.
 - C. 0.95.

Answer = B

The information ratio is calculated by dividing the active return (R_A) by the active risk. The active risk is the standard deviation of the difference between portfolio return and the benchmark return $STD(R_P - R_B)$.

$$IR = \frac{R_P - R_B}{STD(R_P - R_B)} = \frac{R_A}{STD(R_A)}$$

$$IR = \frac{18.54 - 18.14}{1.56} = \frac{0.40}{1.56} = 0.26$$

CFA Level II

"Analysis of Active Portfolio Management," Roger G. Clarke, Harindra de Silva, and Steven Thorley
Section 3.2

57. With respect to the information ratio, which analyst's observation is *least likely* correct?
- A. The observation made by Marano
 - B. The observation made by Gladden
 - C. The observation made by Wert

Answer = A

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The information ratio is unaffected by the aggressiveness of the active weights (deviations from benchmark weights) because both the active return and the active risk increase proportionally.

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Section 3.3

58. In his statement regarding the information and Sharpe ratios, Gladden is *most likely* correct with regard to:
- A. the Sharpe ratio and relative value.
 - B. the impact of cash and leverage.
 - C. absolute versus relative return measures.

Answer = C

The information ratio is a measure of relative expected or realized reward to risk, whereas the Sharpe ratio measures the absolute risk–return trade-off of a portfolio.

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"Analysis of Active Portfolio Management," Roger G. Clarke, Harindra de Silva, and Steven Thorley

Sections 3.1, 3.2

59. Is Wert correct in her assessment of the Fundamental Law of Active Management?
- A. No, she is incorrect about the manager's forecasts
 - B. Yes
 - C. No, she is incorrect about the manager's security weightings

Answer = B

Both observations are correct. A manager's forecasts must correspond at least somewhat to the realized returns if the manager is to generate a positive relative return (information coefficient), and the manager must overweight securities for which he has forecasted the best relative returns in order to generate positive relative returns (transfer coefficient).

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Section 4.1

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60. Is Robbin correct with respect to her assertion about Kariba overstating its expected active return?

- A. No, she is incorrect regarding the impact of the overlap in individual security evaluations
- B. Yes
- C. No, she is incorrect regarding the impact of the number of independent decisions

Answer = B

The expected active return is expressed as the expected information ratio multiplied by the active risk target, or

$$\text{Expected active return} = (\text{IC})\sqrt{\text{BR}} \times (\text{Active risk target})$$

If the number of independent decisions is lower, or if individual securities are affected by an assumption or forecast that persists through multiple rebalancing periods, then breadth will be lower, reducing the information ratio and thus the expected active return.

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Section 6.2

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